


46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA IIT - JEE ACADEMY - INDIA

**Sec: SR.IIT_*CO-SC(MODEL-A& APEX)** **GTM-24 [FN]** **Date: 23.01.2025**
**Time: 3.00Hrs.** **JEE MAIN MODEL** **Max. Marks: 300**

## PHYSICS MAX.MARKS: 100

### SECTION – I
### (SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
**Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.**

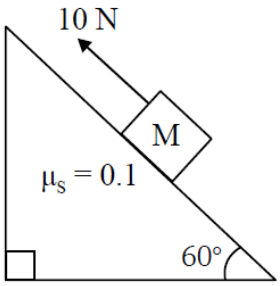
1. If 50 Vernier divisions are equal to 49 main scale divisions of a travelling microscope and one smallest reading of main scale is 0.5 mm, the Vernier constant of travelling microscope is:

A) 0.1 mm B) 0.1 cm C) 0.01 cm D) 0.01 mm



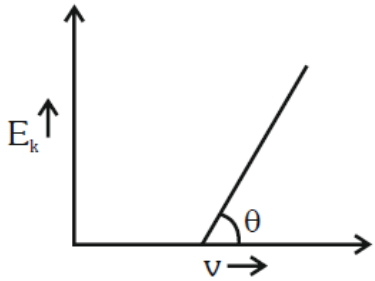
APEX JEE-MAIN 2025

2. A block of mass 1 kg is pushed up a surface inclined to horizontal at an angle of $60°$ by a force of 10 N parallel to the inclined surface as shown in figure. When the block is pushed up by 10 m along inclined surface, the work done against frictional force is: [g=10 $m/s^2$ ]




A) $5\sqrt{3}\,J$ B) $5\,J$ C) $5\times10^3\ J$ D) $10\ J$

3. For the photoelectric effect, the maximum kinetic energy $(E_k)$ of the photoelectrons is plotted against the frequency $(v)$ of the incident photons as shown in figure. The slope of the graph gives



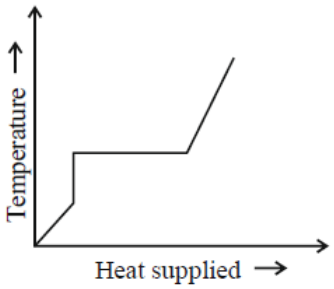

A) Ratio of Planck's constant to electric charge

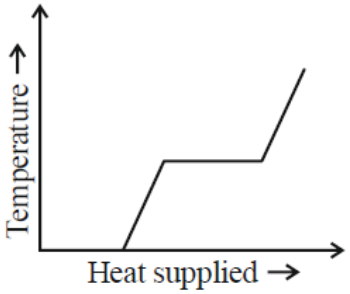
B) Work function of the metal

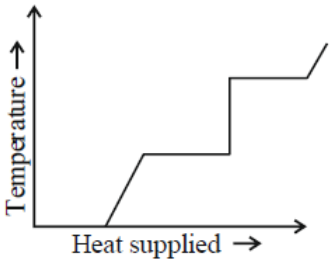
C) Charge of electron

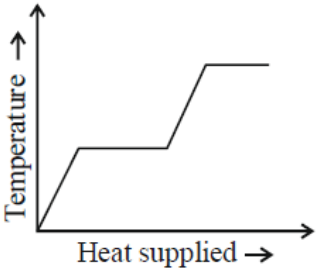
D) Planck's constant

4. A block of ice at $-10°C$ is slowly heated and converted to steam at $100°C$. Which of the following curves represent the phenomenon qualitatively:

A)



B)



C)



D)



5. In a nuclear fission reaction of an isotope of mass M, three similar daughter nuclei of same mass are formed. The speed of a daughter nuclei in terms of mass defect $\Delta M$ will be:

A) $\sqrt{\frac{2c\Delta M}{M}}$ B) $\frac{\Delta Mc^2}{3}$ C) $c\sqrt{\frac{2\Delta M}{M}}$ D) $c\sqrt{\frac{3\Delta M}{M}}$

6. Two resistance of $100\Omega$ and $200\Omega$ are connected in series with a battery of $4V$ and negligible internal resistance. A voltmeter is used to measure voltage across $100\,\Omega$ resistance, which gives reading as $1V$. The resistance of voltmeter must be _____ $\Omega$.

A) 150 B) 100 C) 50 D) 200

7. An electron revolving in $n^{th}$ Bohr orbit has magnetic moment $\mu_n$. If $\mu_n \propto n^x$, the value of $x$ is

A) 2 B) 1 C) 3 D) 0

8. An alternating voltage $V(t) = 220\sin 100\pi t$ volt is applied to a purely resistive load of $50\Omega$. The time taken for the current to rise from half of the peak value to the peak value is:

A) 5 ms B) 3.3 ms C) 7.2 ms D) 2.2 ms



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9. A block of mass $m$ is placed on a surface having vertical cross section given by $y = x^2/4$. If coefficient of friction is 0.5, the maximum height above the ground at which block can be placed without slipping is:

A) 1//4 m B) 1/2 m C) 1/6 m D) 1/3 m

10. If the total energy transferred to a surface in time $t$ is $6.48\times10^5\ J$, then the magnitude of the total momentum delivered to this surface for complete absorption will be:

A) $2.46\times10^{-3}\ kg\ m/s$ B) $2.16\times10^{-3}\ kg\ m/s$

C) $1.58\times10^{-3}\ kg\ m/s$ D) $4.32\times10^{-3}\ kg\ m/s$

11. A beam of unpolarized light of intensity $I_0$ is passed through a polaroid $A$ and then through another polaroid $B$ which is oriented so that its principal plane makes an angle of $45°$ relative to that of $A$. The intensity of emergent light is:

A) $I_0/4$ B) $I_0$ C) $I_0/2$ D) $I_0/8$

12. Escape velocity of a body form earth is 11.2 km/s. If the radius of a planet be one-third the radius of earth and mass be one-sixth that of earth, the escape velocity from the planet is:

A) 11.2 km/s B) 8.4 km/s C) 4.2 km/s D) 7.9 km/s

13. A particle of charge '$-q$' and mass '$m$' moves in a circle of radius '$r$' around an infinitely long line charge of linear density '$+\lambda$'. Then time period will be given as: (Consider $k$ as Coulomb's constant)

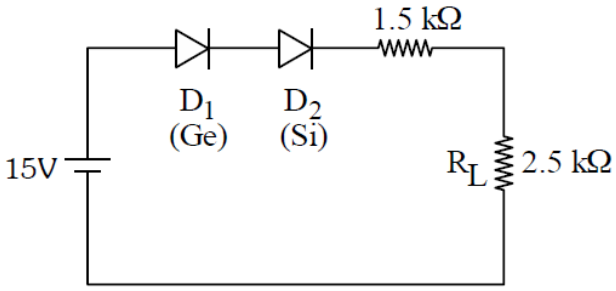
A) $T_2 = \dfrac{4\pi^2 m}{2k\lambda q} r^3$ B) $T = 2\pi r\sqrt{\dfrac{m}{2k\lambda q}}$

C) $T = \dfrac{1}{2\pi r}\sqrt{\dfrac{m}{2k\lambda q}}$ D) $T = \dfrac{1}{2\pi}\sqrt{\dfrac{2k\lambda q}{m}}$

14. If mass is written as $m = kc^P G^{-1/2} h^{1/2}$ then the value of $P$ will be: (Constants have their usual meaning with $k$ a dimensionless constant)

A) 1/2 B) 1/3 C) 2 D) $-1/3$

15. In the given circuit, the voltage across load resistance $(R_L)$ is:

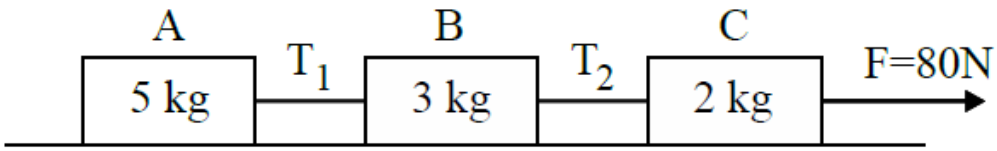



A) 8.75 V B) 9.00 V C) 8.50 V D) 14.00 V

16. If three moles of monoatomic gas $\left(\gamma = \frac{5}{3}\right)$ is mixed with two moles of a diatomic gas $\left(\gamma = \frac{7}{5}\right)$, the value of adiabatic exponent $\gamma$ for the mixture is:

A) 1.75 B) 1.40 C) 1.52 D) 1.35

17. Three blocks A, B and C are pulled on a horizontal smooth surface by a force of 80 N as shown in figure




The tensions $T_1$ and $T_2$ in the string are respectively:

A) 40N, 64N B) 60N, 80N

C) 88N, 96 N D) 80N, 100N

18. When a potential difference V is applied across a wire of resistance R, it dissipates energy at a rate W. If the wire is cut into two halves and these halves are connected mutually parallel across the same supply, the energy dissipation rate will become:

A) 1/4 W B) 1/2 W C) 2 W D) 4 W



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19. Match List I with List II

| | **List-I** | | **List-II** |
|---|---|---|---|
| a. | Gauss's law of magnetostatics | I. | $\oint \vec{E}.\vec{da} = \frac{1}{\varepsilon_0}\int \rho dV$ |
| b. | Faraday's law of electro magnetic induction | II. | $\oint \vec{B}.\vec{da} = 0$ |
| c. | Ampere's law | III. | $\oint \vec{E}.\vec{dl} = \frac{-d}{dt}\int \vec{B}.\vec{da}$ |
| d. | Gauss's law of electrostatics | IV. | $\oint \vec{B}.\vec{dl} = -\mu_0 I$ |

Choose the correct answer from the options given below:

A) $a-I, b-III, c-IV, d-II$ B) $a-III, b-IV, c-I, d-II$

C) $a-IV, b-II, c-III, d-I$ D) $a-II, b-III, c-IV, d-I$

20. A simple pendulum is placed at a place where its distance from the earth's surface is equal to the radius of the earth. If the length of the string is 4m, then the time period of small oscillations will be _____ s. [take: $g = \pi^2 ms^{-2}$]

A) 6 B) 8 C) 12 D) 10

## SECTION-II
## (NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. A power transmission line feeds input power at 2.3 kV to a step down transformer with its primary winding having 3000 turns. The output power is delivered at 230 V by the transformer. The current in the primary of the transformer is 5A and its



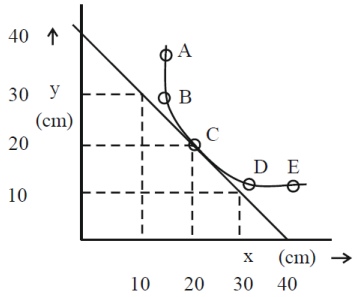
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efficiency is 90%. The winding of transformer is made of copper. The output current of transformer is ____A.

22. A big drop is formed by coalescing 1000 small identical drops of water. If $E_1$ be the total surface energy of 1000 small drops of water and $E_2$ be the surface energy of single big drop of water, the $E_1 : E_2$ is $x:1$ where $x =$ _____

23. Two discs of moment of inertia $I_1 = 4kg\ m^2$ and $I_2 = 2\ kg\,m^2$ about their central axes & normal to their planes, rotating with angular speeds 10 rad/s & 4 rad/s respectively are bought into contact face to face with their axis of rotation coincident. The loss in kinetic energy of the system in the process is _____J.

24. In an experiment to measure the focal length $(f)$ of a convex lens, the magnitude of object distance $(x)$ and the image distance $(y)$ are measured with reference to the focal point of the lens. The $y - x$ plot is shown in figure. The focal length of the lens is _____ cm.




25. The current of $5A$ flows in a square loop of sides $1m$ is placed in air. The magnetic field at the centre of the loop is $X\sqrt{2}\times10^{-7}T$. The value of $X$ is _____.



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**CHEMISTRY** **MAX.MARKS: 100**

**SECTION – I**
**(SINGLE CORRECT ANSWER TYPE)**

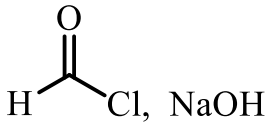
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
**Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.**

26. Which among the following purification methods is based on the principle of "Solubility" in two different solvents?

A) Column Chromatography B) Sublimation

C) Distillation D) Differential Extraction

27. Salicylaldehyde is synthesized from phenol, when reacted with

A)  B) $CO_2, NaOH$

C) $CCl_4$, $NaOH$ D) $HCCl_3$,$NaOH$

28. Given below are two statements:

**Statement – I:** High concentration of strong nucleophilic reagent with secondary alkyl halides which do not have bulky substituents will follow $S_N2$ mechanism.

**Statement –** II: A secondary alkyl halide when treated with a large excess of ethanol follows $S_N1$ mechanism.

In the light of the above statements, choose the most appropriate from the questions given below:

A) Statement I is true but Statement II is false.

B) Statement I is false but Statement II is true.

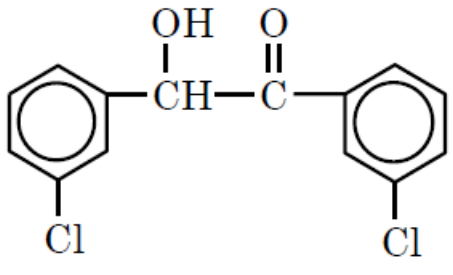
C) Both statement I and Statement II are false.

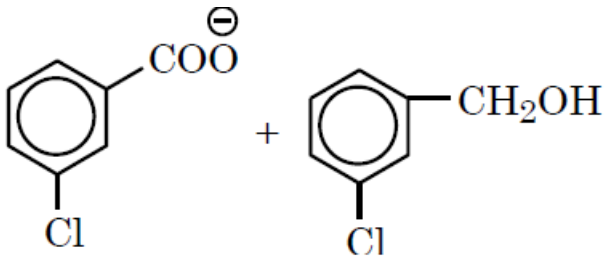
D) Both statement I and Statement II are true.

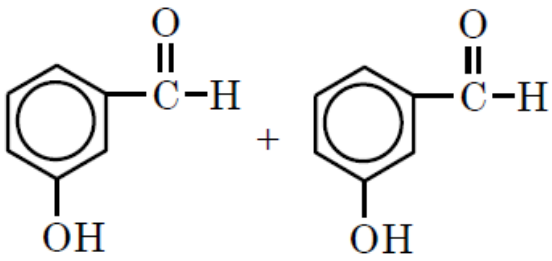


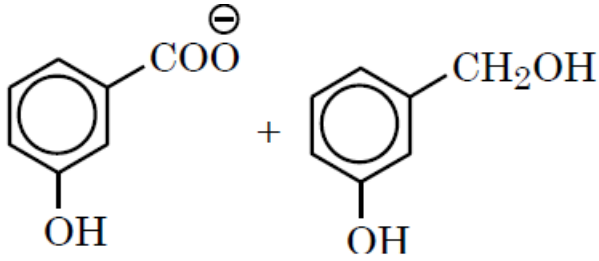
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29. m–chlorobenzaldehyde on treatment with 50% KOH solution yields

A) 

B) 

C) 

D) 

30. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R.

**Assertion A :** $H_2Te$ is more acidic than $H_2S$.

**Reason R:** Bond dissociation enthalpy of $H_2Te$ is lower than $H_2S$.

In the light of the above statements. Choose the most appropriate from the options given below.

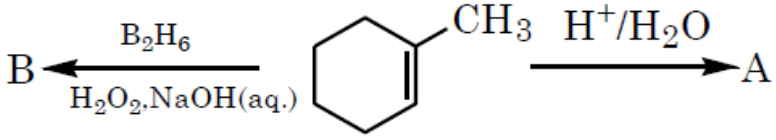
A) Both A and R are true but R is NOT the correct explanation of A.

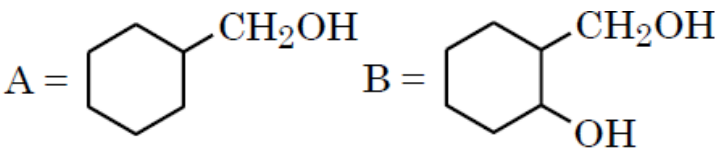
B) Both A and R are true and R is the correct explanation of A.

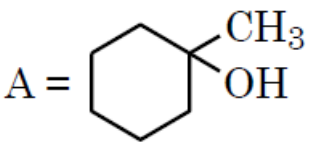
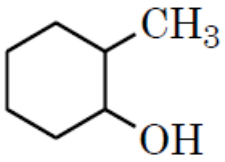
C) A is false but R is true.

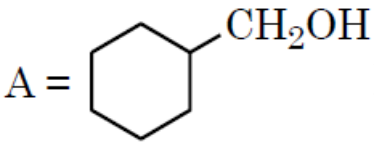
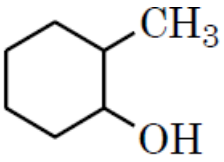
D) A is true but R is false.

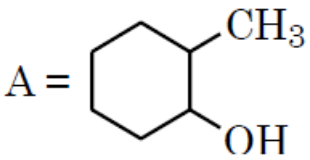
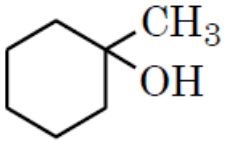
31. Product A and B formed in the following set of reactions are:



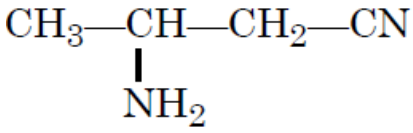
A) 

B) A =  B = 

C) A =  B = 

D) A =  B = 

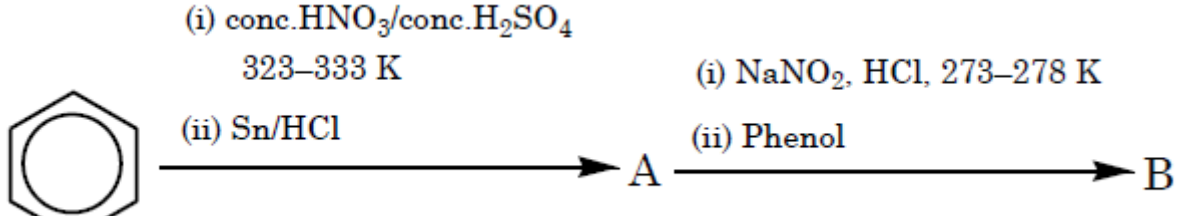
32. IUPAC name of following compound is

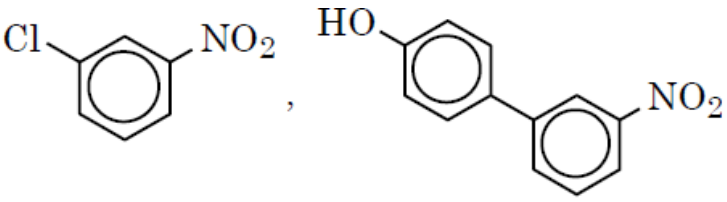


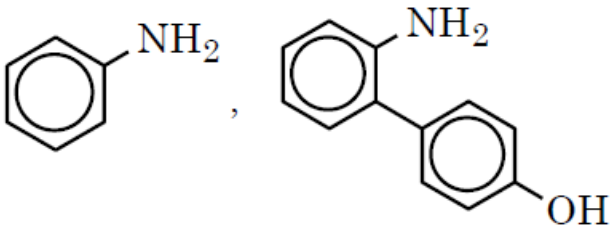
A) 2–Aminopentanenitrile B) 2–Aminobutanenitrile

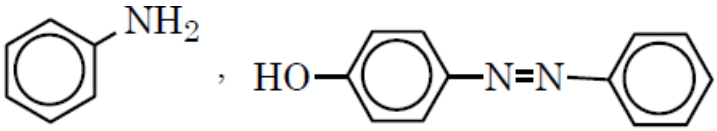
C) 3–Aminobutanenitrile D) 3–Aminopropanenitrile

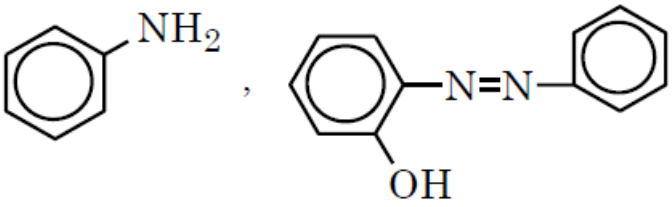
33. The products A and B formed in the following reaction scheme are respectively




A) 

B) 

C) 

D) 

34. The molecule/ion with square pyramidal shape is:

A) $\left[Ni(CN)_4\right]^{2-}$ B) $PCl_5$ C) $BrF_5$ D) $PF_5$

35. The orange color of $K_2Cr_2O_7$ and purple colour of $KMnO_4$ is due to

A) Charge transfer transition in both

B) $d \rightarrow d$ transition in $KMnO_4$ and charge transfer transitions in $K_2Cr_2O_7$.

C) $d \rightarrow d$ transition in $K_2Cr_2O_7$ and charge transfer transitions in $KMnO_4$

D) $d \rightarrow d$ transition in both

36. Alkaline oxidative fusion of $MnO_2$ gives '$A$' which on electrolytic oxidation in alkaline solution produces $B$. $A$ and $B$ respectively are:

A) $Mn_2O_7$ and $MnO_4^-$ B) $MnO_4^{2-}$ and $MnO_4^-$

C) $Mn_2O_3$ and $MnO_4^{2-}$ D) $MnO_4^{2-}$ and $Mn_2O_7$

37. If a substance '$A$' dissolves in solution of a mixture of '$B$' and '$C$' with their respective number of moles as $n_A, n_B$ and $n_C$, mole fraction of $C$ in the solution is"

A) $\frac{n_C}{n_A \times n_B \times n_C}$ B) $\frac{n_C}{n_A + n_B + n_C}$ C) $\frac{n_C}{n_A - n_B - n_C}$ D) $\frac{n_B}{n_A + n_B}$

38. Given below are two statements:

**Statement – I:** Along the period, the chemical reactivity of the element gradually increases from group 1 to group 18.

**Statement – II:** The nature of oxides formed by group 1 element is basic while that of group 17 elements is acidic.

In the the light above statements, choose the most appropriate from the questions given below:

A) Both statement I and Statement II are true.

B) Statement I is true but Statement II is False.

C) Statement I is false but Statement II is true.

D) Both Statement I and Statement II is false.

39. The coordination geometry around the manganese in decacarbonyldimanganese(0)

A) Octahedral

B) Trigonal bipyramidal

C) Square pyramidal

D) Square planar

40. Given below are two statements:

**Statement-I:** Since fluorine is more electronegative than nitrogen, the net dipole moment of $NF_3$ is greater than $NH_3$.

**Statement-II:** In $NH_3$, the orbital dipole due to lone pair and the dipole moment of NH bonds are in opposite direction, but in $NF_3$ the orbital dipole due to lone pair and dipole moments of $N-F$ bonds are in same direction.

In the light of the above statements. Choose the most appropriate from the options given below.

A) Statement I is true but Statement II is false.

B) Both Statement I and Statement II are false.

C) Both statement I and Statement II is are true.

D) Statement I is false but Statement II is are true.

41. The correct stability order of carbocations is

A) $(CH_3)_3C^+ > CH_3-\overset{+}{C}H_2 > (CH_3)_2\overset{+}{C}H > \overset{+}{C}H_3$

B) $\overset{+}{C}H_3 > (CH_3)_2\overset{+}{C}H > CH_3-\overset{+}{C}H_2 > (CH_3)_3\overset{+}{C}$

C) $(CH_3)_3\overset{+}{C} > (CH_3)_2\overset{+}{C}H > CH_3-\overset{+}{C}H_2 > \overset{+}{C}H_3$

D) $\overset{+}{C}H_3 > CH_3-\overset{+}{C}H_2 > CH_3-\underset{\displaystyle CH_3}{\overset{+}{\underset{|}{C}}}H > (CH_3)_3C^+$

42. The solution from the following with highest depression in freezing point/lowest freezing point is

A) 180 g of acetic acid dissolved in water

B) 180 g of acetic acid dissolved in benzene

C) 180 g of benzoic acid dissolved in benzene

D) 180 g of glucose dissolved in water

43. A and B formed in the following reactions are:

$CrO_2Cl_2 + 4NaOH \rightarrow A + 2NaCl + 2H_2O$

$A + 2HCl + 2H_2O_2 \rightarrow B + 3H_2O$

A) $A = Na_2CrO_4, B = CrO_5$ B) $A = Na_2Cr_2O_4, B = CrO_4$

C) $A = Na_2Cr_2O_7, B = CrO_3$ D) $A = Na_2Cr_2O_7, B = CrO_5$

44. Choose the correct statements about the hydrides of group 15 elements.

a. The stability of the hydrides decreases in the order

$NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$

b. The reducing ability of the hydrides increases in the order

$NH_3 < PH_3 < AsH_3 < SbH_3 < BiH_3$

c. Among the hydrides, $NH_3$ is strong reducing agent while $BiH_3$ is mild reducing agent.

d. The basicity of the hydrides increases in the order

$NH_3 < PH_3 < AsH_3 < SbH_3 < BiH_3$

Choose the most appropriate from the option given below:

A) b and c only

B) c and d only

C) a and b only

D) a and d only

45. Reduction potential of ions of given below:

| $ClO_4^-$ | $IO_4^-$ | $BrO_4^-$ |
|---|---|---|
| $E^\circ = 1.19V$ | $E^\circ = 1.65V$ | $E^\circ = 1.74V$ |

The correct order of their oxidizing power is:

A) $ClO_4^- > IO_4^- > BrO_4^-$

B) $BrO_4^- > IO_4^- > ClO_4^-$

C) $BrO_4^- > ClO_4^- > IO_4^-$

D) $IO_4^- > BrO_4^- > ClO_4^-$

## SECTION-II
## (NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. Number of complexes which show optical isomerism among the following is _____

cis- $\left[Cr(ox)_2Cl_2\right]^{3-}$, $\left[Co(en)_3\right]^{3+}$, cis-$\left[Pt(en)_2Cl_2\right]^{2+}$, cis-$\left[Co(en)_2Cl_2\right]^{+}$,

trans-$\left[Pt(en)_2Cl_2\right]^{2+}$, trans-$\left[Cr(ox)_2Cl_2\right]^{3-}$

47. $NO_2$ required for a reaction is produced by decomposition of $N_2O_5$ in $CCl_4$ as by equation $2N_2O_{5(g)} \rightarrow 4NO_{2(g)} + O_{2(g)}$. The initial concentration of $N_2O_5$ is $3\ mol\ L^{-1}$



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and it is $2.75\ mol\ L^{-1}$ after 30 minutes. The rate of formation of $NO_2$ is $x \times 10^{-3}\ mol\ L^{-1}\ \min^{-1}$, value of $x$ is _____

48. Two reactions are given below:

$$2Fe_{(s)} + \frac{3}{2}O_{2(g)} \rightarrow Fe_2O_{3(s)}, \Delta H^\circ = -822\,kJ\ /\ mol$$

$$C_{(s)} + \frac{1}{2}O_{2(g)} \rightarrow CO_{(g)}, \Delta H^\circ = -110\,kJ\ /\ mol$$

Then enthalpy change for following reaction

$$3C_{(s)} + Fe_2O_{3(s)} \rightarrow 2Fe_{(s)} + 3CO_{(g)}$$

49. The total number of correct statements, regarding the nucleic acids is_____.

A. RNA is regarded as the reserve of genetic information.

B. DNA molecule self-duplicates during cell division

C. DNA synthesizes proteins in the cell.

D. The message for the synthesis of particular proteins is present in DNA

E. Identical DNA strands are transferred to daughter cells.

50. The pH of an aqueous solution containing 1M benzoic acid $\left(pK_a = 4.20\right)$ and $1M$ sodium benzoate is 4.5. The volume of benzoic acid solution is $300\ ml$ of this buffer solution is _____ $ml$.

**MATHS** **MAX.MARKS: 100**

**SECTION – I**
**(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
**Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.**

51. Consider the system of linear equations

$x+y+z=5,\ x+2y+\lambda^2 z=9,$

$x+3y+\lambda z=\mu,$ where $\lambda,\mu\in R$. Then, which of the following statement is NOT correct?

A) System has infinite number of solution if $\lambda=1$ and $\mu=13$

B) System is inconsistent if $\lambda=1$ and $\mu\neq 13$

C) System is consistent if $\lambda\neq 1$ and $\mu=13$

D) System has unique solution if $\lambda\neq 1$ and $\mu\neq 13$

52. For $\alpha,\beta\in\left(0,\frac{\pi}{2}\right)$, let $3\sin(\alpha+\beta)=2\sin(\alpha-\beta)$ and a real number $k$ be such that $\tan\alpha=k\tan\beta$. Then the value of $k$ is equal to:

A) $-\frac{2}{3}$ B) $-5$ C) $\frac{2}{3}$ D) $5$

53. Let $A(\alpha,0)$ and $B(0,\beta)$ be the points on the line $5x+7y=50$. Let the point $P$ divide the line segment $AB$ internally in the ratio $7:3$. Let $3x-25=0$ be a directrix of the ellipse $E:\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and the corresponding focus be $S$. If from $S$, the perpendicular on the $x$-axis passes through $P$, then the length of the latus rectum of $E$ is equal to

A) $\frac{25}{3}$ B) $\frac{32}{9}$ C) $\frac{25}{9}$ D) $\frac{32}{5}$



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54. Let $\vec{a}=\hat{i}+\alpha\hat{j}+\beta\hat{k}, \alpha,\beta\in R$. Let a vector $\vec{b}$ be such that the angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{4}$ and $\left|\vec{b}\right|^2=6$. If $\vec{a}.\vec{b}=3\sqrt{2}$, then the value of $\left(\alpha^2+\beta^2\right)\left|\vec{a}\times\vec{b}\right|^2$ is equal to

A) 90 B) 75 C) 95 D) 85

55. Let $f\left(x\right)=\left(x+3\right)^2\left(x-2\right)^3, x\in\left[-4,4\right]$. If $M$ and $m$ are the maximum and minimum values of $f$, respectively in $\left[-4,4\right]$, then the value of $M-m$ is:

A) 600 B) 392 C) 608 D) 108

56. Let $a$ and $b$ be two distinct positive real numbers. Let 11$^{th}$ term of a GP, whose first term is $a$ and third term is $b$, is equal to $p^{th}$ term of another GP, whose first term is $a$ and fifth term is $b$. Then $p$ is equal to

A) 20 B) 25 C) 21 D) 24

57. If $x^2-y^2+2hxy+2gx+2fy+c=0$ is the locus of a point, which moves such that it is always equidistant from the lines $x+2y+7=0$ and $2x-y+8=0$, then the value of $g+c+h-f$ equals

A) 14 B) 6 C) 8 D) 29

58. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $\left|\vec{b}\right|=1$ and $\left|\vec{b}\times\vec{a}\right|=2$. Then $\left|\left(\vec{b}\times\vec{a}\right)-\vec{b}\right|^2$ is equal to

A) 3 B) 5 C) 1 D) 4

59. Let $y=f\left(x\right)$ be a thrice differentiable function in $\left(-5,5\right)$. Let the tangents to the curve $y=f\left(x\right)$ at $\left(1,f\left(1\right)\right)$ and $\left(3,f\left(3\right)\right)$ makes angles $\frac{\pi}{6}$ and $\frac{\pi}{4}$, respectively with positive $x$-axis. If $27\int_1^3\left(f'\left(t\right)^2+1\right)f''\left(t\right)dt=\alpha+\beta\sqrt{3}$ where $\alpha,\beta$ are integers, then the value of $\alpha+\beta$ equals

A) –14 B) 26 C) –16 D) 36



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60. Let $P$ be a point on the hyperbola $H:\frac{x^2}{9}-\frac{y^2}{4}=1$, in the first quadrant such that the area of the triangle formed by $P$ and the two foci of $H$ is $2\sqrt{13}$. Then, the square of the distance of $P$ from the origin is

A) 18 B) 26 C) 22 D) 20

61. Bag A contains 3 white, 7 red balls and bag B contains 3 white, 2 red balls. One bag is selected at random and a ball is drawn from it. The probability of drawing the ball from the bag A, if the ball drawn in white, is:

A) $\frac{1}{4}$ B) $\frac{1}{9}$ C) $\frac{1}{3}$ D) $\frac{3}{10}$

62. Let $f:R\to R$ be defined as $f(x)=ae^{2x}+be^{x}+cx$. If $f(0)=-1, f'(\log_e 2)=21$ and $\int_0^{\log_e 4}\left(f(x)-cx\right)dx=\frac{39}{2}$, then the value of $|a+b+c|$ equals:

A) 16 B) 10 C) 12 D) 8

63. Let $L_1:\vec{r}=\left(\hat{i}-\hat{j}+2\hat{k}\right)+\lambda\left(\hat{i}-\hat{j}+2\hat{k}\right),\lambda\in R$ $L_2:\vec{r}=\left(\hat{j}-\hat{k}\right)+\mu\left(3\hat{i}+\hat{j}+p\hat{k}\right),\mu\in R$ and $L_3:\vec{r}=\delta\left(l\hat{i}+m\hat{j}+n\hat{k}\right),\delta\in R$ be three lines such that $L_1$ is perpendicular to $L_2$ and $L_3$ is perpendicular to both $L_1$ and $L_2$. Then the point which lies on $L_3$ is

A) $(-1,7,4)$ B) $(-1,-7,4)$ C) $(1,7,-4)$ D) $(1,-7,4)$

64. Let $a$ and $b$ be real constant such that the function $f$ defined by

$f(x)=\begin{cases}x^2+3x+a, & x\le 1\\ bx+2, & x>1\end{cases}$ be differentiable on $R$. Then, the value of $\int_{-2}^{2} f(x)dx$ equals

A) $\frac{15}{6}$ B) $\frac{19}{6}$ C) 21 D) 17



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65. Let $f:R-\{0\}\to R$ be a function satisfying $f\left(\frac{x}{y}\right)=\frac{f(x)}{f(y)}$ for all $x,y, f(y)\neq 0$. If $f'(1)=2024$, then

A) $xf'(x)-2024f(x)=0$ B) $xf'(x)+2024f(x)=0$

C) $xf'(x)+f(x)=2024$ D) $xf'(x)-2023f(x)=0$

66. If $z$ is a complex number, then the number of common roots of the equation $z^{1985}+z^{100}+1=0$ and $z^3+2z^2+2z+1=0$, is equal to:

A) 1 B) 2 C) 0 D) 3

67. Let $Y=Y(X)$ be a curve lying in the first quadrant such that the area enclosed by the line $Y-y=Y'(x)(X-x)$ and the co-ordinate axes, where $(x,y)$ is any point on the curve, is always $\frac{-y^2}{2Y'(x)}+1, Y'(x)\neq 0$. If $Y(1)=1$, then $Y(2)$ equals _____.

A) $\frac{5}{3}$ B) 2 C) $\frac{1}{3}$ D) 1

68. If the domain of the function $f(x)=\log_e\left(\frac{2x+3}{4x^2+x-3}\right)+\cos^{-1}\left(\frac{2x-1}{x+2}\right)$ is $(\alpha,\beta]$, then the value of $5\beta-4\alpha$ is equal to

A) 10 B) 12 C) 11 D)9

69. Let $f:R\to R$ be a function defined $f(x)=\frac{x}{\left(1+x^4\right)^{1/4}}$ and $g(x)=f\left(f\left(f\left(f(x)\right)\right)\right)$ then $18\int_0^{\sqrt{2\sqrt{5}}}x^2g(x)dx$

A) 33 B) 36 C) 42 D) 39

70. Let $R=\begin{pmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{pmatrix}$ be a non-zero $3\times3$ matrix, where

$x\sin\theta = y\sin\left(\theta+\frac{2\pi}{3}\right) = z\sin\left(\theta+\frac{4\pi}{3}\right)\neq 0,\ \theta\in(0,2\pi)$. For a square matrix $M$, let trace $(M)$ denote the sum of all the diagonal entries of $M$. Then among the statements:

(I) Trace $(R)=0$

(II) If trace $\left(adj\left(adj(R)\right)\right)=0$, then $R$ has exactly one non-zero entry.

A) Both (I) and (II) are true

B) Neither (I) nor (II) is true

C) Only (II) is true

D) One (I) is true

## SECTION-II
## (NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. Consider two circles $C_1: x^2+y^2=25$ and $C_2:(x-\alpha)^2+y^2=16$, where $\alpha\in(5,9)$. Let the angle between the two radii (one to each circle) drawn from one of the intersection points of $C_1$ and $C_2$ be $\sin^{-1}\left(\frac{\sqrt{63}}{8}\right)$. If the length of common chord of $C_1$ and $C_2$ is $\beta$, then the value of $(\alpha\beta)^2$ equals _____.



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72. Let $\alpha=\sum_{k=0}^{n}\left(\frac{\left({}^{n}C_{k}\right)^{2}}{k+1}\right)$ and $\beta=\sum_{k=0}^{n-1}\left(\frac{{}^{n}C_{k}\,{}^{n}C_{k+1}}{k+2}\right)$. If $5\alpha=6\beta$, then $n$ equals _____.

73. Let $S_n$ be the sum to $n$-terms of an arithmetic progression 3, 7, 11, ….. .

If $40<\left(\frac{6}{n(n+1)}\sum_{k=1}^{n}S_{k}\right)<42$, then $n$ equals _____ .

74. The number of symmetric relations defined on the set $\{1,2,3,4\}$ which are not reflexive is _____

75. The area of the region enclosed by the parabola $(y-2)^2=x-1$, the line $x-2y+4=0$ and the positive coordinate axes is _____ .